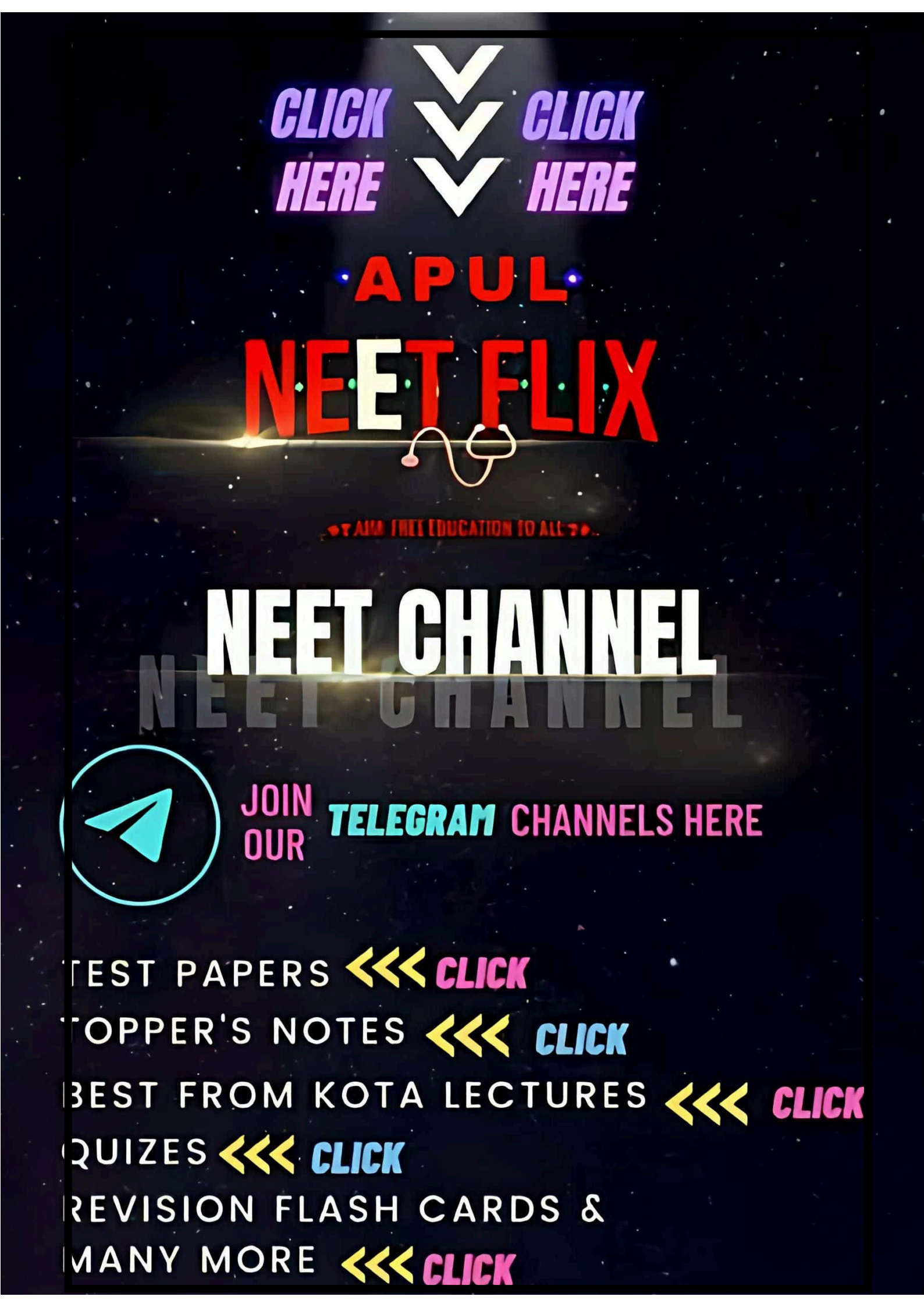

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Some Basic Concepts of Chemistry - Mole Concept

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1. Which statement is correct ?
(1) Solids have definite volume and definite shape
(2) Liquids have definite volume but do not have definite shape
(3) Gases have neither definite volume nor definite shape
(4) All of these
CMC253

2. A mixture has :
(1) Two or more pure substances in any ratio
(2) Two types (a) homogeneous mixture
(b) heterogeneous mixture
(3) Properties somewhat similar to its components
(4) All of these
CMC254

3. A pure substance :
(1) Has fixed composition of the constituent particles.
(2) Has two types : element & compound, out of which element consists of only one type of atom and compound consists of more than one type of atoms combined together in a definite ratio by mass
(3) Cannot be separated into its constituents by simple physical methods.
(4) All of these
CMC255

4. One amu is :
(1) Having mass exactly equal to one-twelfth of the mass of one carbon-12 atom
(2) 1.66×10^{-24}g
(3) Also known as unified mass (u)
(4) All of these
CMC256

5. For NaCl 58.5 amu is :
(1) Its molas mass (2) Its formula mass
(3) Its molecular mass (4) Its atomic mass
CMC257

6. What mass of water can be produced by combustion of 16 g methane ?
(1) 36 g (2) 18 g
(3) 54 g (4) 8 g
CMC258

7. How many moles of methane are required to produce 66 g of CO_2 after combustion ?
(1) 1 mol (2) 1.25 mol
(3) 1.50 mol (4) 0.325 mol
CMC259

8. A solution is prepared by adding 2 g of solute P to 38 g of water. Calculate the mass percent of the solute P.
(1) 10% (2) 5%
(3) 1% (4) 4%
CMC260

9. Which statement is correct ?
(1) The properties of compound are altogether different from those of its constituent elements
(2) The SI system has seven base unit
(3) The SI unit of volume is litre
(4) Both (1) and (2)
CMC261

10. The boiling point of water is :
(1) 100°C (2) 373 K
(3) 212° F (4) All of these
CMC262

11. Which is incorrect according to Dalton's atomic theory ?
(1) Matter consists of indivisible atoms.
(2) All atoms of a given element have identical properties, including identical mass.
(3) Compounds are formed when atoms of different elements combine in a indefinite ratio.
(4) Chemical reaction involves reorganization of atoms. These atoms are neither created nor destroyed in a chemical reaction.
CMC263

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12. which is the correct SI unit ?

Column- I		Column-II	
(P)	Volume	(a)	kg/m^3
(Q)	Density	(b)	m^3
(R)	Temperature	(c)	candela
(S)	Luminous intensity	(d)	kelvin

(1) P → b, Q → a, R → d, S → c
(2) P → b, Q → a, R → c, S → d
(3) P → c, Q → b, R → a, S → d
(4) P → d, Q → a, R → b, S → c

CMC264

13. Find the mass % of hydrogen in ethanol (C_2H_5OH).
(1) 13.13% (2) 34.73%
(3) 52.14% (4) 23.10%

CMC265

14. An element has two isotopes having atomic mass 10 u and 15 u respectively. If the % abundance of heavier isotope is 20% then the average atomic mass of the element is :
(1) 9 u (2) 11 u (3) 12 u (4) 14 u

CMC266

15. An aqueous solution contains 2 g of solute in 16 g of water. The mass % of solute is :
(1) 11.1% (2) 10%
(3) 12.89% (4) 60.45%

CMC267

16. The density of 3M solution of NaCl is 1.25 g/mL. Calculate the molality of the solution.
(1) 1.79 mol/kg (2) 3.79 mol/kg
(3) 2.79 mol/kg (4) 4.79 mol/kg

CMC268

17. If 2 mol of N_2 and 3 mol H_2 are mixed together to produce NH_3 then select the incorrect option :
(1) H_2 is the limiting reagent
(2) N_2 is the limiting reagent
(3) The amount of NH_3 is governed by H_2
(4) All are correct

CMC269

18. 2M, 4L aqueous HCl solution is mixed with 6L water, the molarity of the resultant solution is:
(1) 1.68 M (2) 0.80 M
(3) 1.40 M (4) 0.40 M

CMC270

19. Equal volume of all gases at the same temperature and pressure should contain equal number of molecules.
This statement is in accordance with:
(1) Avogadro's law
(2) Law of multiple proportion
(3) Law of conservation of mass
(4) Gay Lussac's law of gaseous volumes

CMC271

20. Concentration term which changes with temperature:-
(1) Mass percent (2) Molality
(3) Molarity (4) All of these

CMC272

21. For the reaction
$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$
(a) One mole of $CH_{4(g)}$ reacts with two moles of $O_{2(g)}$ to give one mole of $CO_{2(g)}$ and two moles of $H_2O_{(g)}$
(b) One molecule of $CH_{4(g)}$ reacts with 2 molecules of $O_{2(g)}$ to give one molecule of $CO_{2(g)}$ and 2 molecules of $H_2O_{(g)}$
(c) 22.7 litre of $CH_{4(g)}$ reacts with 45.4L of $O_{2(g)}$ to give 22.7 L of $CO_{2(g)}$ and 45.4 L of $H_2O_{(g)}$
(d) 16g of $CH_{4(g)}$ reacts with 64g of O_2 to give 44g of $CO_{2(g)}$ and 36g of $H_2O_{(g)}$
Which option is correct ?
(1) Only a and b (2) Only a, b and d
(3) Only a, b and c (4) All are correct

CMC273

22. Ratio of amount of CO_2 produced when 1 mole of carbon is burnt in air and also in 16 g of dioxygen is:
(1) 1 : 2 (2) 2 : 1
(3) 1 : 1 (4) 3 : 1

CMC274

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23. How much copper can be obtained from 79.75 g of $CuSO_4$?
(1) 31.75 g (2) 63.5 g
(3) 79.75 g (4) 100 g
CMC275

24. In three moles of ethane which of the following is correct?
(1) 6 gram atom of carbon is present
(2) 18 gram atom of hydrogen is present
(3) $3N_A$ molecules of C_2H_6 is present
(4) All of these
CMC276

25. If ten volume of dihydrogen gas reacts with five volumes of dioxygen gas, how many volumes of water vapour would be produced?
(1) 100 volume (2) 10 volume
(3) 1000 volume (4) 1 volume
CMC277

26. Which one of the following will have the largest no. of atoms ?
(1) 1g Al(s) (2) 1g Na(s)
(3) 1g Li(s) (4) 1g Cl_2(g)
CMC278

27. Calculate the ratio of number of atoms in 52 mol Ar and 52 g He ?
(1) 4 : 1 (2) 1 : 4 (3) 1 : 1 (4) 2 : 1
CMC279

28. Calculate the mass of sodium acetate required to make 500mL of 0.4 molar aqueous solution.
(Molar mass of sodium acetate is 82 g mol^{-1})
(1) 16.4 g (2) 15.1 g
(3) 14.3 g (4) 41.2 g
CMC280

29. 1 dm^3 is equal to.
(1) 1 L (2) 1000 cc
(3) 10^{-3} m^3 (4) All of these
CMC281

30. Statement-I:- Mole(symbol mol) was introduced as seventh base quantity for amount of substance in SI system.
Statement-II :- 1 mole contains $6.02214076 \times 10^{23}$ elementary entities.
(1) Statement I is correct but statement II is incorrect
(2) Statement I and Statement II both are correct
(3) Statement I and Statement II both are incorrect
(4) Statement I is incorrect but statement II is correct
CMC282

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**4**	**4**	**4**	**2**	**1**	**3**	**2**	**4**	**4**	**3**	**1**	**1**	**2**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**2**	**1**	**3**	**4**	**2**	**1**	**4**	**2**	**3**	**1**	**1**	**4**	**2**

Atomic Structure

1. Calculate the sum of number of protons, neutrons and electrons in $^{80}_{35}Br$.

(1) 45 (2) 80 (3) 85 (4) 115

CAS316

2. Calculate frequency of yellow radiation having wavelength 5800Å

(1) $5.172 \times 10^{18}\ s^{-1}$ (2) $5.172 \times 10^{14}\ s^{-1}$

(3) $5.172 \times 10^{19}\ s^{-1}$ (4) $5.172 \times 10^{16}\ s^{-1}$

CAS317

3. Calculate energy of one mole of photons of radiation whose frequency is 5×10^{14} Hz.

(1) 199.51 J mol^{-1} (2) 199.51 kJ/mol

(3) 199.51×10^{3} kJ/mol (4) None of these

CAS318

4. What will be the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 ms^{-1} ?

(1) 6.626×10^{34} m (2) 6.626×10^{36} m

(3) 6.626×10^{-34} m (4) 6.626×10^{-36} m

CAS319

5. Calculate the mass of a photon with wavelength 3.313 Å.

(1) 1.5×10^{-29} kg (2) 0.66×10^{-30} kg

(3) 6.6×10^{-29} kg (4) 6.6×10^{-33} kg

CAS320

6. A golf ball has a mass of 10g, and a speed of 50m/s. If the speed can be measured with an accuracy of 2%, calculate the uncertainty in the position.

(1) 0.527×10^{-34} m (2) 5.27×10^{-33} m

(3) 5.27×10^{-34} m (4) 52.7×10^{-34} m

CAS321

7. Match the following species with their corresponding ground state electronic configuration.

Atom/Ion		Electronic configuration	
(i)	Cu	(a)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$
(ii)	Cu^{2+}	(b)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2$
(iii)	Zn^{+2}	(c)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
(iv)	Cr^{+3}	(d)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$
		(e)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$

(1) (i) → (c), (ii) → (d), (iii) → (a), (iv) → (e)

(2) (i) → (b), (ii) → (d), (iii) → (a), (iv) → (e)

(3) (i) → (c), (ii) → (d), (iii) → (a)(b), (iv) → (e)

(4) (i) → (b), (ii) → (e)(d), (iii) → (a), (iv) → (c)

CAS322

8. Assertion (A):- It is impossible to determine the exact position and exact momentum of an electron simultaneously.

Reason (R):- The path of an electron in an atom is clearly defined.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is not the correct explanation of A.

(3) A is true and R is false.

(4) Both A and R are false.

CAS323

9. In which of the following pairs, the ions are isoelectronic ?

(1) Na^+, Mg^{2+} (2) Al^{3+}, O^-

(3) Na^+, O (4) N^{3-}, Cl^-

CAS324

10. The pair of ions having same electronic configuration is :

(1) Cr^{3+}, Fe^{3+} (2) Fe^{3+}, Mn^{2+}

(3) Fe^{3+}, Co^{3+} (4) Sc^{3+}, Cr^{3+}

CAS325

11. What is the number of photons of light with a wavelength of 4000pm that provide 1J of energy ?

(1) 2×10^{17} (2) 2×10^{16}

(3) 2×10^{19} (4) 2×10^{15}

CAS326

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12. Match the items of column I and column II configuration.

Column-I		Column-II	
(i)	r_n	(a)	$\propto \frac{z^2}{n^3}$
(ii)	V_n	(b)	$\propto \frac{n^2}{z}$
(iii)	T_n	(c)	$\propto \frac{z}{n}$
(iv)	ν_n	(d)	$\propto \frac{n^3}{z^2}$

(1) (i) → (b), (ii) → (c), (iii) → (d), (iv) → (a)
(2) (i) → (b), (ii) → (d), (iii) → (a), (iv) → (c)
(3) (i) → (c), (ii) → (d), (iii) → (b), (iv) → (a)
(4) (i) → (a), (ii) → (d), (iii) → (c), (iv) → (b)

CAS327

13. Two atoms are said to be isodiaphers if.
(1) They have same atomic number but different mass number
(2) They have same number of neutrons but different number of electrons
(3) They have same difference of the number of neutrons and protons.
(4) Sum of the number of protons and neutrons is same but the number of protons is different.

CAS328

14. What is the atomic number of the element whose outermost electronic configuration is $3d^6$?
(1) 27 (2) 25
(3) 26 (4) 24

CAS329

15. The number of angular nodes and radial nodes in 4f orbital are
(1) 3 and 0, respectively
(2) 4 and 0, respectively
(3) 0 and 3, respectively
(4) 0 and 4, respectively

CAS330

16. Which of the following statement is not correct about the characteristics of cathode rays ?
(1) They start from the cathode and move towards the anode
(2) They travel in straight line in the absence of an external electrical or magnetic field.
(3) Characteristics of cathode rays do not depend upon the material of electrodes in cathode ray tube.
(4) Characteristics of cathode rays depend upon the nature of gas present in the cathode ray tube.

CAS331

17. Which of the following statements concerning the quantum numbers are correct?
(1) Angular quantum number does not determine the three dimensional shape of the orbital.
(2) The principal quantum number determines the orientation and energy of the orbital.
(3) Magnetic quantum number determines the size of the orbital.
(4) Spin quantum number of an electron determines the orientation of the spin of electron relative to the chosen axis.

CAS332

18. How many electrons in an atom may have the following quantum numbers n = 4, $m_s = -\frac{1}{2}$?
(1) 2 (2) 14 (3) 16 (4) 10

CAS333

19. Which is a correct statement ?
(1) The volume of the nucleus is about 10^{15} times the volume of the atom.
(2) The volume of the nucleus is about 10^{10} times the volume of the atom.
(3) The volume of the atom is about 10^{15} times the volume of the nucleus.
(4) The volume of the atom is about 10^{10} times the volume of the nucleus.

CAS334

20. Which is not correct ?

(1) On increasing radius, total energy of e^- increases.

(2) On increasing radius, P.E. of e^- decreases.

(3) On increasing radius, K.E. of e^- decreases.

(4) On increasing radius, P.E. of e^- Increases.

CAS335

21. Which is not a correct relationship ?

(1) $\Delta x.\ \Delta p \geq \frac{h}{4\pi}$ (2) $\Delta x.\ \Delta v \geq \frac{h}{4\pi m}$

(3) $\Delta E \times \Delta t \geq \frac{h}{4\pi.\Delta x}$ (4) $F \times \Delta t \times \Delta x \geq \frac{h}{4\pi}$

CAS336

22. Which is an incorrect statement ?

(1) Isotopes do not have the same value of e/m.

(2) Isobars do not have the same value of e/m.

(3) Isotopes do not have same chemical property but they have different physical property.

(4) Isobars do not have the same chemical and physical property.

CAS337

23. Which is correct ?

(1) The diameter (size) of the atom is 10^{15} times the diameter of the nucleus.

(2) The diameter (size) of the nucleus is 10^{15} times the diameter of the atom.

(3) The diameter (size) of the nucleus is 10^5 times the diameter of the atom.

(4) The diameter (size) of the atom is 10^5 times the diameter of the nucleus.

CAS338

24. Which of the following sets of quantum numbers is possible ?

(1) $n = 0, l = 0, m_\ell = 0, m_s = +\frac{1}{2}$

(2) $n = 1, l = 1, m_\ell = 0, m_s = -\frac{1}{2}$

(3) $n = 2, l = 1, m_\ell = 0, m_s = -\frac{1}{2}$

(4) $n = 1, l = 1, m_\ell = 0, m_s = +\frac{1}{2}$

CAS339

25. Calculate the number of electrons in the species, H_2^+, H_2 and O_2^+ respectively.

(1) 0, 2, 15

(2) 1, 2, 15

(3) 1, 2, 16

(4) 0, 2, 16

CAS340

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**2**	**3**	**4**	**2**	**1**	**3**	**1**	**2**	**2**	**1**	**3**	**3**	**1**
Question	16	17	18	19	20	21	22	23	24	25					
Answer	**4**	**4**	**3**	**3**	**2**	**3**	**3**	**4**	**3**	**2**					

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Redox Reactions

1. In the reaction given below, Identify the species undergoing oxidation and reduction.

$3Fe_3O_{4(s)} + 8Al_{(s)} \longrightarrow 9Fe_{(s)} + 4Al_2O_{3(s)}$

(1) Al, Fe_3O_4 (2) Fe, Fe_3O_4

(3) Fe_3O_4, Al (4) Al_2O_3, Fe

CRR191

2. What is/are oxidising agents ?

(1) Acceptor of electrons

(2) Donor of electrons

(3) Both (1) and (2)

(4) None of these

CRR192

3. $Zn_{(s)} + Cu^{+2}{}_{(aq)} \longrightarrow Zn^{+2}{}_{(aq)} + Cu_{(s)}$

Which of the following will act as oxidising agent in the above reaction ?

(1) Zn (2) Cu^{+2}

(3) Zn^{+2} (4) Cu

CRR193

4. Which is the correct stock notation to represent the compound $HAuCl_4$?

(1) HAu(III)Cl (2) HAu(I)Cl

(3) HAu(II)Cl (4) HAu(IV)Cl

CRR194

5. $2Cu_2O_{(g)} + Cu_2S_{(g)} \longrightarrow 6Cu_{(s)} + SO_{2(g)}$

Identify the species which act as reducing agent.

(1) Cu_2O (2) Cu_2S

(3) Cu (4) SO_2

CRR195

6. $TiCl_{4\,(l)} + 2Mg_{\,(s)} \xrightarrow{\Delta} Ti_{(s)} + 2MgCl_{2\,(s)}$

Find oxidation number of Ti in $TiCl_{4(l)}$ and $Ti_{(s)}$

(1) 0, +4 (2) –4, 0

(3) +4, 0 (4) -4, +2

CRR196

7. Which of the following species do not show disproportionation reaction ?

ClO^-, ClO_2^-, ClO_3^- and ClO_4^-

(1) ClO^- (2) ClO_2^-

(3) ClO_3^- (4) ClO_4^-

CRR197

8. How many compounds have fractional oxidation number ?

C_3O_2, Br_3O_8, H_2O, CrO_5, $Na_2S_4O_6$

(1) 2 (2) 4 (3) 3 (4) 5

CRR198

9. $xMnO^-_{4(aq)} + yBr^-_{(aq)} + H_2O_{(l)} \rightarrow zMnO_{2(s)} + BrO^-_{3(aq)} + 2OH^-_{(aq)}$

Permanganate ion reacts with bromide ion in basic medium to give manganese dioxide and bromate ion.

Write the balanced ionic equation for the reaction and determine x, y and z.

(1) 2, 1, 1 (2) 2, 1, 2

(3) 4, 1, 2 (4) 3, 2, 1

CRR199

10. Assign oxidation number to the underlined elements in each of the following species:-

(a) $NaH_2\underline{P}O_4$

(b) $Ca\underline{O}_2$

(c) $H_2S_2\underline{O}_7$

(d) $KAl(\underline{S}O_4)_2 \cdot 12H_2O$

Options are :-

(1) +4, –2, –2, +6 (2) +5, +2, –2, +6

(3) +5, –1, –2, +6 (4) None of these

CRR200

11. Fluorine reacts with ice and results in the change

$H_2O_{(s)} + F_{2\,(g)} \rightarrow HF_{(g)} + HOF_{(g)}$

Calculate O.N. of O in HOF.

(1) 0 (2) 1 (3) –2 (4) –1

CRR201

12. Calculate the oxidation no. of sulphur, chromium and nitrogen in H_2SO_5, $Cr_2O_7^{-2}$ and NO_3^- respectively.

(1) +6, +6, +6 (2) +4, +4, +2

(3) +6, +6, +5 (4) +5, +4, +2

CRR202

13. Write formulae for the following compounds
(a) Mercury (II) Chloride
(b) Nickel (II) Sulphate
(c) Iron (III) Sulphate
(d) Thallium (I) Sulphate
Options are :-
(1) $HgCl, NiSO_4, Fe_3(SO_4)_2, Tl_2SO_4$
(2) $HgCl_2, NiSO_4, Fe_2(SO_4)_3, Tl_2SO_4$
(3) $HgCl_2, NiSO_4, Fe_3(SO_4)_2, TlSO_4$
(4) None of these

CRR203

14. Identify the oxidising agent and reducing agent for following reactions :

$$2AgBr_{(s)} + C_6H_6O_{2(aq)} \rightarrow 2Ag_{(s)} + 2\,HBr_{(aq)} + C_6H_4O_{2(aq)}$$

(1) $AgBr, C_6H_6O_2$ (2) $C_6H_6O_2, AgBr$
(3) $Ag, C_6H_6O_2$ (4) $AgBr, C_6H_4O_2$

CRR204

15. Find the value of x and y for the following reaction

$$xFe^{+2}_{(aq)} + yCr_2O^{2-}_{7\,(aq)} + 14H^{+}_{(aq)} \rightarrow Fe^{+3}_{(aq)} + Cr^{+3}_{(aq)}$$

(1) 6, 1 (2) 5, 2
(3) 1, 6 (4) 2, 5

CRR205

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**1**	**2**	**1**	**2**	**3**	**4**	**3**	**2**	**3**	**1**	**3**	**2**	**1**	**1**

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Thermodynamics

1. A well stoppered thermos flask contains some ice cubes. This is an example of :
(1) Closed system
(2) Open system
(3) Isolated system
(4) Non-thermodynamic system.
CTD373

2. For the reaction.
$C(s) + O_2(g) \rightarrow CO_2(g)$
(1) $\Delta H > \Delta U$ (2) $\Delta H < \Delta U$
(3) $\Delta H = \Delta U$ (4) None of these
CTD374

3. For 1 mol Ideal gas C_P & C_V are related as :
(1) $C_P - C_V = R$ (2) $C_P + C_V = R$
(3) $C_P - C_V = RT$ (4) $C_V - C_P = R$
CTD375

4. Considering entropy (S) as the criteria for the spontaneity of any process is :
(1) $\Delta S_{system} + \Delta S_{surrounding} > 0$
(2) $\Delta S_{system} - \Delta S_{surrounding} < 0$
(3) $\Delta S_{system} > 0$
(4) $\Delta S_{surrounding} > 0$
CTD376

5. If Liquids A & B form an Ideal solution :
(1) The entropy of mixing is zero.
(2) The free energy of mixing is zero.
(3) The free energy as well as the entropy of mixing are zero.
(4) The enthalpy of mixing is zero.
CTD377

6. 3 Litre of an Ideal gas at a pressure of 0.1 atm expands isothermally at 25°C into a vacuum until its total volume is 10L. How much heat is absorbed & how much work is done in the expansion ?
(1) 0.7 (2) 1
(3) 0 (4) None of these
CTD378

7. If water vapour is assumed to be a perfect gas, molar enthalpy change for vaporisations of 1 mole water at 1 bar & 100°C is 41 kJ mol^{-1}, calculate the internal energy change, when 1 mol of water is vapourised at 1 bar pressure & 100°C.
(1) 37.9 J mol^{-1} (2) 37.9 kJ mol^{-1}
(3) 59.7 kJ mol^{-1} (4) 23.4 J mol^{-1}
CTD379

8. The enthalpy & entropy changes for the reaction :
$Br_2(\ell) + Cl_2(g) \longrightarrow 2BrCl(g)$
Are 30 kJ mol^{-1} & 105 Jk^{-1} mol^{-1} respectively. The temperature at which the reaction will be in equilibrium is :
(1) 285.7 K (2) 273.4 K
(3) 450.9 K (4) 300.1 K
CTD380

9. For the reaction.
$2Br(g) \longrightarrow Br_2(g)$ the value of ΔH & ΔS respectively are :
(1) $\Delta H = 0$; ΔS = –ve
(2) $\Delta H = 0$; $\Delta S = 0$
(3) ΔH = –ve ; ΔS = –ve
(4) ΔH = +ve ; ΔS = +ve
CTD381

10. Change in entropy is negative for :
(1) $Br(\ell) \longrightarrow Br(g)$
(2) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$
(3) $N_2(g)$ (10 atm) $\longrightarrow N_2(g)$ (1 atm)
(4) Fe (400K) $\longrightarrow$ Fe (300K)
CTD382

11. Which among the following is not a state function ?
(1) Internal energy (2) Free energy
(3) Work (4) Enthalpy
CTD383

12. The enthalpy of formation of all elements in their standard state is:
(1) Unity
(2) Zero
(3) Less than zero
(4) Different for each element
CTD384

13. Thermodynamics is not concerned about :
(1) Energy changes involved in a chemical reaction.
(2) The extent to which a chemical reaction proceeds.
(3) The rate at which a reaction proceeds
(4) The feasibility of a chemical reaction
CTD385

14. Which of the following is open system:
(1) Animal & plants (2) A fridge
(3) A solar cooker (4) None of these
CTD386

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15. The work done during expansion of a gas from a volume of $2dm^3$ to $4dm^3$ against a constant external pressure of 1 atm is :
(1) –202.6 J (2) + 202.6 J
(3) –304 J (4) –6 J
CTD387

16. In adiabatic process, which of the following is true ?
(1) q = +w (2) $-\Delta E = -w$
(3) $p\Delta U = 0$ (4) $q = \Delta E$
CTD388

17. In an isochoric process, the increase in internal energy is:
(1) Equal to the heat absorbed
(2) Equal to heat evolved
(3) Equal to the work done
(4) Equal to zero
CTD389

18. At constant temperature for the reaction :
$C_4H_{10(g)} + \frac{13}{2} O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(\ell)}$
$\Delta E - \Delta H$ is ?
(1) + RT (2) – RT
(3) $+\frac{7}{2}$ RT (4) –3RT
CTD390

19. When a liquid vaporizes ;
(1) No increase in enthalpy
(2) Increase in enthalpy
(3) Decrease in enthalpy
(4) Anything can happen
CTD391

20. For the given reaction :
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ $\Delta H = 4kJ$
The ΔH is specifically called :
(1) Heat of formation of NH_3
(2) Heat of combustion
(3) Heat of reaction
(4) Heat of Hydrogenation
CTD392

21. Given below are two statements :
Statement-I : natural processes are irreversible & spontaneous process.
Statement-II : Decrease in enthalpy is a contributory factor for spontaneity.
(1) Both statement-I and II are incorrect.
(2) Statement-I is correct & statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I & II are correct.
CTD393

22. Which of the following correct option for free expansion of an ideal gas under adiabatic condition ?
(1) q = 0 ; $\Delta T \neq 0$; w = 0
(2) $q \neq 0$; $\Delta T = 0$; w = 0
(3) q = 0 ; $\Delta T = 0$; w = 0
(4) q = 0 ; $\Delta T < 0$; $w \neq 0$
CTD394

23. Bond energy of a molecule :
(1) Is always positive
(2) Is always negative
(3) Either positive or negative
(4) Depends upon the physical state of the system.
CTD395

24. If $\Delta G° < 0$ for a reaction, then :
(1) $K_P > 1$
(2) $K_P < 1$
(3) The product dominates in the equilibrium mixture
(4) None
CTD396

25. Entropy means measurement of degree of :
(1) Disordness
(2) Randomness
(3) Orderness
(4) Both (1) and (2)
CTD397

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	1	4	3	2	1	3	4	3	2	3	1	1
Question	16	17	18	19	20	21	22	23	24	25					
Answer	2	1	3	2	3	4	3	2	1	4					

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Chemical Equilibrium

1. **Assertion (A) :-** When Argon is added at constant volume, equilibrium remains undisturbed.
Reason (R) :- The addition of inert gas at constant volume does not change the partial pressure or the molar concentrations of the substance involved in the reaction.
(1) A and R both are correct and R is correct explanation of A
(2) A and R both are correct but R is not the correct explanation of A
(3) A is correct but R is incorrect
(4) Both A and R are incorrect
CCE214

2. Which among the following factor disturbs the state of equilibrium but doesn't alter the value of K ?
(1) Addition of Krypton at constant volume
(2) Increasing the concentration of reactant
(3) Increasing temperature
(4) All of these
CCE215

3. Which of the following is correct at equilibrium condition ?
(1) $\ln K = -\Delta G°/RT$ (2) $K = e^{-\Delta G°/RT}$
(3) $\Delta G = 0$ (4) All of these
CCE216

4. Which is correct feature of physical equilibrium :

Process	Condition
(1) $H_2O_{(\ell)} \rightleftharpoons H_2O_{(g)}$	$P_{H2O(g)}$ is constant at given temperature
(2) $H_2O_{(s)} \rightleftharpoons H_2O_{(\ell)}$	Melting point is fixed at constant pressure
(3) $Sugar_{(s)} \rightleftharpoons Sugar_{(aq)}$	Concentration of solute in solution is constant at a given temperature

(4) All are correct
CCE217

5. Le chatelier's principle is applicable for.
(1) Physical equilibrium
(2) Chemical equilibrium
(3) Both physical and chemical equilibrium
(4) None of these
CCE218

6. At 450K, $K_p = 2 \times 10^{10}$ bar^{-1} for the reaction ;
$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$
What is K_C at this temperature ?
(1) 7.47×10^{11} M^{-1} (2) 7.47×10^{9} M^{-1}
(3) 7.47×10^{8} M^{-1} (4) 7.47×10^{13} M^{-1}
CCE219

7. For the reaction :
$Fe^{+3}_{(aq)} + SCN^{-}_{(aq)} \rightleftharpoons [Fe(SCN)]^{+2}_{(aq)}$
(yellow) (colourless) (deep red)
Which statement is correct ?
(1) Addition of oxalic acid will decrease the intensity of red colour
(2) Addition of $HgCl_2$ decreases the intensity of red colour
(3) Addition of potassium thiocyanate increases the intensity of red colour
(4) All of these
CCE220

8. Hydrolysis of sucrose gives
Sucrose + $H_2O \rightleftharpoons$ Glucose + Fructose
Equilibrium constant K_C for the reaction is 2×10^{13} at 27°C, calculate the value of $\Delta G°$ at 27°C
(1) -7.64×10^{4} J mol^{-1} (2) -7.64×10^{3} J mol^{-1}
(3) -8.64×10^{30} J mol^{-1} (4) -7.64×10^{2} J mol^{-1}
CCE221

9. For :
$NO_{(g)} + O_{3(g)} \rightleftharpoons NO_{2(g)} + O_{2(g)}$; $K_C = 6.3 \times 10^{14}$
Find K_C for reverse reaction.
(1) 2×10^{14} (2) 1.59×10^{-15}
(3) 1.6×10^{-10} (4) 1.6×10^{-20}
CCE222

10. Equilibrium plays a vital role in our body in:
(1) Maintaining pH of blood.
(2) Transportation of O_2 from lungs to muscle.
(3) Both (1) and (2)
(4) None of these

CCE223

11. Which represents characteristics of equilibrium:-
(1) It is dynamic in nature
(2) All measurable properties remain constant
(3) A catalyst doesn't affect the state of equilibrium
(4) All of these

CCE224

12. At equilibrium, the concentrations of N_2, O_2 and NO in a sealed vessel at 800K are 3×10^{-3} M, 4.2×10^{-3} M and 2.8×10^{-3} M respectively. What will be the K_C for the reaction:
$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$
(1) 0.622
(2) 0.311
(3) 0.287
(4) 0.234

CCE225

13. For the equilibrium
$2NOCl_{(g)} \rightleftharpoons 2NO_{(g)} + Cl_{2(g)}$
The value of equilibrium constant, K_C is 3.75×10^{-6} at 1069 K. Calculate the K_P for the reaction at this temperature ?
(1) 0.00066 (2) 0.00033
(3) 0.00022 (4) 0.00011

CCE226

14. Which is correct statement/statements ?
(1) If $K_C > 10^3$, products predominate over reactants.
(2) If $K_C < 10^{-3}$, reactants predominate over products
(3) If K_C is in the range of 10^{-3} to 10^3, appreciable concentrations of both reactant and product are present
(4) All of these

CCE227

15. For the reaction :
$2A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$, If K_C for the reaction is 3×10^{-3} and at a given time, the composition of reaction mixture is
$[A] = [B] = [C] = 3 \times 10^{-4}$M.
In which direction the net reaction will proceed ?
(1) Forward
(2) Backward
(3) Neither forward nor backward
(4) All of these

CCE228

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	4	4	3	1	4	1	2	3	4	1	2	4	2

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Ionic Equilibrium

1. Which statement is correct ?

(1) The electrostatic forces between two charges are inversely proportional to dielectric constant of medium

(2) Water is universal solvent and possesses a very high dielectric constant.

(3) When sodium chloride is dissolved in water the electrostatic interactions are reduced by a factor of 80 and this facilitates the ions to move freely in solution

(4) All of these

CIE363

2. $HCl_{(aq)} + H_2O_{(\ell)} \rightleftharpoons H_3O^+_{(aq)} + Cl^-_{(aq)}$

Column-I		Column-II	
(a)	Conjugate base of HCl	(p)	HCl
(b)	Conjugate acid of Cl^-	(q)	Cl^-
(c)	Conjugate base of H_3O^+	(r)	H_3O^+
(d)	Conjugate acid of H_2O	(s)	H_2O

(1) a → q, b → p, c → s, d → r

(2) a → p, b → q, c → r, d → s

(3) a → s, b → r, c → q, d → p

(4) None of these

CIE364

3. Which option is wrong ?

	Species	conjugate acid	conjugate base
(1)	H_2O	H_3O^+	OH^-
(2)	HCO_3^-	H_2CO_3	CO_3^{-2}
(3)	NH_3	NH_4^+	NH_2^-
(4)	NH_3	NH_2^-	NH_4^+

CIE365

4. Which is always true for a neutral solution?

(1) pH = 7

(2) $[H_3O^+] = [OH^-]$

(3) pH = pOH

(4) Both (2) and (3)

CIE366

5. If pH increases by 2 units, the (H^+)

(1) Decreases to 100 times

(2) Decreases to 10 times

(3) Increases to 10 times

(4) Increases to 100 times

CIE367

6. The concentration of hydrogen ion in a sample of soft drink is 3.8×10^{-3} M. What is its pH ?

(1) 1.42

(2) 3.8

(3) 3–log 3.8

(4) Both (1) and (3)

CIE368

7. Calculate the pOH of 1.0×10^{-8} M solution of HCl, at 25°C

(1) 6.98 (2) 7.02

(3) 8.00 (4) 6.00

CIE369

8.

	Acid	K_a
(a)	HF	3.5×10^{-4}
(b)	HNO_2	4.5×10^{-4}
(c)	HCOOH	1.8×10^{-4}
(d)	CH_3COOH	1.74×10^{-5}

Increasing order of acidic strength will be:

(1) a > b > c > d (2) b > a > c > d

(3) d < c < b < a (4) c < b < d < a

CIE370

9. The pk_a of acetic acid and pK_b of ammonium hydroxide are 4.76 and 4.75 respectively. Calculate the pH of aqueous solution of ammonium acetate.

(1) 7.005 (2) 7.000

(3) 6.995 (4) 8.445

CIE371

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10. Which option is correct ?

Salt	solubility (s)
(a) soluble	S > 0.1 M
(b) slightly soluble	0.01 M < S < 0.1 M
(c) sparingly soluble	S < 0.01 M

(1) only a
(2) only a & b
(3) only b & c
(4) All are correct

CIE372

11. K_{SP} of $Zr_3(PO_4)_4$ is :
(1) 6912 s^7 (2) 6942 s^1
(3) 6912 s^{27} (4) 6921 s^4

CIE373

12. The value of K_{SP} of two sparingly soluble salts $Ni(OH)_2$ and AgCN are 2.0×10^{-15} and 6×10^{-17} respectively. Which salt is more soluble ?
(1) AgCN
(2) $Ni(OH)_2$
(3) Both have same solubility
(4) Nothing can be predicted.

CIE374

13. Calculate the solubility of A_2X_3 in pure water, assuming that neither kind of ion reacts with water. The solubility product, K_{SP} of A_2X_3 is 1.1×10^{-23}.
(1) 1.0×10^{-5} mol/L
(2) 2.0×10^{-5} mol/L
(3) 3.0×10^{-5} mol/L
(4) 4.0×10^{-5} mol/L

CIE375

14. Find the molar solubility of $Ni(OH)_2$ in 0.10 M NaOH if K_{SP} of $Ni(OH)_2$ is 2.0×10^{-15}.
(1) 2.0×10^{-13}
(2) 2.0×10^{-12}
(3) 2.0×10^{-11}
(4) 2.0×10^{-10}

CIE376

15. A 0.02 M solution of pyridinium hydrochloride has pH = 3.44. Find the ionization constant of pyridine.
(1) 1.5×10^{-5}
(2) 1.5×10^{-9}
(3) 1.5×10^{-4}
(4) 1.5×10^{-2}

CIE377

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	4	4	1	3	2	2	1	4	1	2	1	1	2

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Electrochemistry

1. Write Nernst equation at equilibrium condition:-

(1) $\frac{2.303nF}{RT}\log K_{eq}$

(2) $2.303\ RT\log K_{eq}$

(3) $\frac{2.303RT}{nF}\log K_{eq}$

(4) None of these

CEL230

2. For the cell in which the following reaction takes place

$Mg_{(s)} + 2Ag^{+}(0.0001M) \rightarrow Mg^{2+}(0.130M) + 2Ag_{(s)}$

Calculate its $E_{(cell)}$ if E^{0}_{cell} = 3.17 V.

(1) 2.14 V (2) 2.96 V

(3) 3.01 V (4) 2.80 V

CEL231

3. Electrons flow from Cu to Zn in Daniell cell when :-

(1) $E_{ext} < 1.1V$ (2) $E_{ext} > 1.1$

(3) $E_{ext} = 1.1V$ (4) All of these

CEL232

4. What is electrode Potential for standard hydrogen electrode at all temperature ?

(1) 0 (2) +1.1

(3) –1.1 (4) 2.4

CEL233

5. What does the negative value of E^{0} represents ?

(1) Strong reducing agent then H^{+}/H_2 couple

(2) Strong oxidising agent then H^{+}/H_2 couple

(3) Weak reducing agent then H^{+}/H_2 couple

(4) None of these

CEL234

6. Calculate the equilibrium constant of the reaction :

$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

$E^{0}_{(cell)} = 0.46V$

(1) 3.19×10^{16} (2) 3.92×10^{15}

(3) 4.01×10^{11} (4) 3.92×10^{17}

CEL235

7. The standard electrode potential for Daniel cell is 1.1V. calculate the standard Gibbs energy for the reaction :

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

(1) –219 KJ/mol (2) –212.27 KJ/mol

(3) –201.92 KJ/mol (4) –192 KJ/mol

CEL236

8. The value of specific conductance and molar conductance on dilution respectively :-

(1) Increases, Decreases

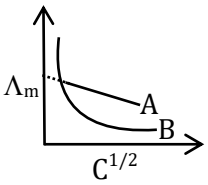
(2) Decreases, Increases

(3) Decreases, Decreases

(4) Increases, Increases

CEL237

9.



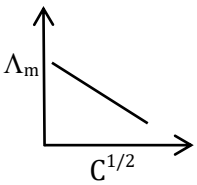
A and B respectively are :-

(1) HCN, NaCl (2) KCl, NaCl

(3) KCl, CH_3COOH (4) CH_3COOH, HCN

CEL238

10. For which of the following electrolyte limiting molar conductance can be obtained by the following graph-



(1) KCl (2) CH_3COOH

(3) HCN (4) NH_4OH

CEL239

11. Limiting molar conductivity of an electrolyte can be represented as :-

(1) Cations individual contribution

(2) Anions individual contribution

(3) Sum of cations and anions individual contribution

(4) None of these

CEL240

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12. Λ_m^0 for NaCl, HCl and NaAc are 126.4, 425.9 and 91.0 S cm^2 mol^{-1} respectively. Calculate Λ^0 for HAc.

(1) 295.6 S cm^2 mol^{-1} (2) 385.2 S cm^2 mol^{-1}
(3) 390.5 S cm^2 mol^{-1} (4) 405.28 S cm^2 mol^{-1}

CEL241

13. The conductivity of 0.001028 mol L^{-1} acetic acid is 4.95×10^{-5} S cm^{-1}. Calculate its dissociation constant if Λ_m^0 for acetic acid is 390.5 S cm^2 mol^{-1}.

(1) 1.68×10^{-4} mol L^{-1} (2) 1.86×10^{-5} mol L^{-1}
(3) 4.56×10^{-6} mol L^{-1} (4) 1.78×10^{-5} mol L^{-1}

CEL242

14. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode?

(1) 0.38 g (2) 0.456 g
(3) 0.2938 g (4) 0.598 g

CEL243

15. Galvanic cells that are designed to convent the energy of combustion of methane into electrical energy are called :-

(1) Primary cell (2) Secondary cell
(3) Fuel cell (4) Mercury cell

CEL244

16. In which of the following cell reactants are continuously supplied from external source ?

(1) Primary cells
(2) Fuel cells
(3) Lead storage battery
(4) Mercury cell

CEL245

17. Arrange the following metals in the order in which they displace each other from the solution of their salts.

Al, Cu, Fe, Mg and Zn

(1) Al < Cu < Fe < Mg < Zn
(2) Cu < Fe < Zn < Al < Mg
(3) Mg < Cu < Fe < Zn < Al
(4) Fe < Zn < Cu < Al < Mg

CEL246

18. Given the standard electrode potentials,

$K^+/K = -2.93V$, $Ag^+/Ag = 0.80V$,

$Hg^{2+}/Hg = 0.79V$

$Mg^{2+}/Mg = -2.37$ V, $Cr^{3+}/Cr = -0.74V$

Arrange these metals in their increasing order of reducing power.

(1) K > Cr > Hg > Ag > Mg
(2) K > Hg > Ag > Cr > Mg
(3) K > Mg > Cr > Hg > Ag
(4) K > Mg > Cr > Ag > Hg

CEL247

19. Calculate the emf with the help of Nernst equation at 298 K:

$Mg(s) | Mg^{2+}(0.001M) || Cu^{2+}(0.0001\ M) | Cu(s)$

$E^0_{Mg/Mg^{2+}} = 2.37\,V$

$E^0_{Cu/Cu^{2+}} = -0.34\,V$

(1) 5.75 V (2) 2.68 V
(3) 1.68 V (4) 3.42 V

CEL248

20. In the button cells widely used in watches and other devices the following reaction takes place:

$Zn(s) + Ag_2O(s) + H_2O(l) \rightarrow Zn^{2+}(aq) + 2Ag(s) + 2OH^-(aq)$

Determine $\Delta_r G^0$ and E^0 for the reaction.

$E^0_{Ag^+/Ag} = 0.80\,V$

$E^0_{H_2O/OH^-} = -0.46\,V$

$E^0_{Zn/Zn^{2+}} = 0.76\,V$

(1) 2.057 V & –513 KJ
(2) 1.104 V & –214 KJ
(3) 4.057 V & –613 KJ
(4) none of the above

CEL249

21. How much charge is required for the reduction of 1 mol of Al^{3+} to Al ?

(1) 6.257×10^7 C (2) 2.895×10^5 C
(3) 2.895×10^7 C (4) none of the above

CEL250

22. How much electricity in terms of Faraday is required to produce 20.0 g of Ca from molten $CaCl_2$?

(1) 1F (2) 2F

(3) 0.5F (4) 3F

CEL251

23. How much electricity is required in coulomb for the oxidation of 1 mol of H_2O to O_2 ?

(1) 2.73×10^5 C (2) 1.93×10^5 C

(3) 2.73×10^7 C (4) 1.93×10^7 C

CEL252

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**2**	**1**	**1**	**2**	**2**	**2**	**3**	**1**	**3**	**3**	**4**	**3**	**3**
Question	16	17	18	19	20	21	22	23							
Answer	**2**	**2**	**3**	**2**	**2**	**2**	**1**	**2**							

Solutions

1. The molarity of 500 mL solution in which 5 g NaOH is dissolved, is :
(1) 0.275 M (2) 0.2 M
(3) 0.25 M (4) 2.5 M
CSO257

2. The density of 20% (w/w) aqueous KI solution is 1.202 g/mL then find out the molality of solution.
(1) 1.25 m (2) 1.5 m
(3) 2 m (4) 0.101 m
CSO258

3. In a solution 39 g benzene is dissolved in 46 g ethyl alcohol then the mole fraction of benzene is :
(1) 0.5 (2) 0.67
(3) 0.33 (4) 0.25
CSO259

4. The concentration term which depends on temperature is :
(1) Mass percentage
(2) Mole fraction
(3) Molarity
(4) Molality
CSO260

5. 2.5 g acetic acid is dissolved in 75 g benzene then molality of solution is :
(1) 0.55 m (2) 0.43 m
(3) 0.25 m (4) 0.4 m
CSO261

6. The solution in which equilibrium is established in between solution and undissolved solute is :
(1) Unsaturated solution
(2) Saturated solution
(3) Hypersaturated solution
(4) True solution
CSO262

7. In commercial bleaching solution the chemical present with 3.62% by weight (w/w) in water is :
(1) $CaOCl_2$ (2) NaOCl
(3) NaOH (4) Cl_2
CSO263

8. The concentration of ethylene glycol solution used for cooling of engines in vehicles is :
(1) 25 % (V/V) (2) 35 % (V/V)
(3) 45 % (V/V) (4) 15 % (V/V)
CSO264

9. What will be the mass of urea to prepare 3.5 L aqueous solution with 0.4 molar concentration?
(1) 38 g (2) 42 g
(3) 168 g (4) 84 g
CSO265

10. On increasing the temperature, the solubility of gas in liquid :
(1) Increases (2) Decreases
(3) Remains constant (4) Can't predicted
CSO266

11. 1g of a non-electrolyte when dissolved in 50g benzene, its freezing point is decreased by 0.40 K. K_f of benzene is 5.12 K kg mol^{-1}. Molar mass of solute will be :-
(1) 128 (2) 256
(3) 512 (4) 64
CSO267

12. During osmosis flow of solvent molecules takes place from :
(1) Pure solvent to solution
(2) Solution to pure solvent
(3) Both (1) and (2)
(4) Flow of solvent molecules does not occur
CSO268

13. If blood cells are placed in solution with less than 0.9% (w/v) salt solution then
(1) They will shrink/collapse
(2) They will burst
(3) They will multiply
(4) None of these
CSO269

14. Desalination of sea water is carried out by the process -
(1) Exosmosis (2) Endosmosis
(3) Reverse osmosis (4) Osmosis
CSO270

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15. What will be the mass of ascorbic acid $(C_6H_8O_6)$ which is dissolved in 75 g acetic acid to decrease its freezing point by 1.5°C. K_f = 3.9 K kg mol^{-1}.
(1) 5.077 g (2) 5.36 g
(3) 10.15 g (4) 0.44 g
CSO271

16. In case of association of solute in solution, Van't Hoff factor is :
(1) 1 (2) Less than 1
(3) More than 1 (4) Infinite
CSO272

17. The values of Van't Hoff factors for KBr, $MgCl_2$ and Na_2SO_4 respectively, are :
(1) 2, 2, 3 (2) 2, 3, 3
(3) 2, 3, 7 (4) 1, 1, 1
CSO273

18. Which of the following aqueous solutions will have the highest boiling point?
(1) 1 M NaOH (2) 1 M $MgSO_4$
(3) 1 M KNO_3 (4) 1 M $CaCl_2$
CSO274

19. From which of the following binary mixture, their components can not be separated completely by fractional distillation.
(1) n-hexane and n-heptane
(2) C_2H_5Br and C_2H_5Cl
(3) C_6H_6 and $C_6H_5–CH_3$
(4) H_2O and HNO_3
CSO275

20. 4 L of 0.02 M aqueous solution of NaCl is diluted with 1 L water then molarity of final solution is :
(1) 0.004 M (2) 0.008 M
(3) 0.016 M (4) 0.02 M
CSO276

21. In comparison to a 0.01 M solution of urea, the elevation in boiling point of a 0.01 M $CaBr_2$ solution is :
(1) The same (2) About twice
(3) About three times (4) About six times
CSO277

22. $K_3[Fe(CN)_6]$ is 50% ionised in solution then which of the following is correct?
(1) i = 2 (2) i = 4
(3) i = 2.5 (4) i = 5.5
CSO278

23. The value of cryoscopic constant depends on :
(1) Number of particles of solute
(2) Number of particles of solvent
(3) Mole fraction of solute
(4) Freezing point of solvent
CSO279

24. Osmotic pressure of solution is proportional to :-
(1) Molarity of solution
(2) Molality of solution
(3) Normality of solution
(4) Formality of solution
CSO280

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	3	3	1	2	1	2	4	2	2	3	1	3	1
Question	16	17	18	19	20	21	22	23	24						
Answer	2	2	4	4	3	3	3	2	1						

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Chemical Kinetics

1. For a first order reaction, $t_{99\%} = y \times t_{99.90\%}$, Find value of y.

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) 3 (4) $\frac{4}{3}$

CCK249

2. For the reaction A→ B, the concentration of reactant changes from 0.05 to 0.03 in 30 sec, calculate average rate of reaction.

(1) 6.6×10^{-4} MS^{-1}

(2) 3.2×10^{-3} MS^{-1}

(3) 0.5×10^{-5} MS^{-1}

(4) 6.3×10^{-7} MS^{-1}

CCK250

3. For a reaction 2A + B → C + 3D, Rate = $K[A]^1[B]^1$ if concentration of A is doubled and B will be $\frac{1}{4}$ times find change in rate ?

(1) doubled (2) half

(3) $\frac{1}{3}$ time (4) $\frac{1}{4}$ time

CCK251

4. It takes 20 min to decompose $BaCO_3$ to $\frac{1}{4}$ of its initial amount, If the decomposition is a 1st order reaction calculate the rate constant of the reaction.

(1) 0.0144 (2) 0.0325

(3) 0.0425 (4) 0.0214

CCK252

5. The value of rate constant of a pseudo 1st order reaction.

(1) Depends on the concentration of reactant present in excess.

(2) Depend on the concentration of reactant present in a small amount.

(3) Independent of concentration of reactant.

(4) Depends only on temperature.

CCK253

6. Write rate equation for the reaction 2A+B→C if order of reaction is zero.

(1) $r=K[A][B]$ (2) $r=K[A]^2[B]$

(3) $r=K[A][B]^2$ (4) $r=K[A]^0[B]^0$

CCK254

7. Can molecularity of any reaction be equal to zero ?

(1) Yes (2) No

(3) Can be Yes or No (4) None of these

CCK255

8. The rate of a catalyst to change

(1) Gibbs energy of reaction

(2) Enthalpy of reaction

(3) Activation energy of reaction

(4) Equation constant

CCK256

9. For reaction A + B → product, Rate Law is given by $r=K[A]^{1/2}[B]^1$, what is order of reaction.

(1) 2 (2) $\frac{3}{2}$ (3) 1 (4) 2.5

CCK257

10. For a 1st order reaction, find the rate constant if half-life is 200 s. :-

(1) $3.4 \times 10^{-3}s^{-1}$ (2) $1.2 \times 10^{-5}s^{-1}$

(3) $4.2 \times 10^{-2}s^{-1}$ (4) $2.7 \times 10^{-2}s^{-1}$

CCK258

11. A 1st order reaction takes 40 min for 30% decomposition calculate $t_{1/2}$ = ?

(1) 84.3 min (2) 75.32 min

(3) 45.3 min (4) 42 min

CCK259

12. The rate constant of a reaction at 500K & 700K are 0.03 s^{-1} and 0.09 s^{-1}. Calculate value of Ea.

(1) 15987.13 (2) 913.2

(3) 459.53 (4) 784.5

CCK260

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13. The rate of reaction quadruples when temperature charges from 273 to 293K. If initial rate is $3.2\times10^{2}MS^{-1}$, final rate is-

(1) 6.4×10^{2} (2) 7.8×10^{2}

(3) 9.5×10^{2} (4) 12.8×10^{2}

CCK261

14. The rate constant for 1st order reaction is $70s^{-1}$. How much time it take to reduce initial concentration of reactant to its $\frac{1}{12^{th}}$ value ?

(1) 0.055 (2) 0.095

(3) 0.035 (4) 0.015

CCK262

15. A reaction is 1st order in A and 2nd order in B. How is the rate affected on increasing the concentration of B three times ?

(1) 6 time (2) 8 time

(3) 9 time (4) 4 time

CCK263

16. A reaction is 1st order in A and 2nd order in B, Write differential rate equation for it.

(1) $r=K[A][B]$

(2) $r=K[A]^{1/2}[B]$

(3) $r=K[A][B]^{2}$

(4) $r=K[A]^{2}[B]$

CCK264

17. The rate constant for 1st order decomposition of H_2O_2 is given by the following equation :-

$\log K = 15-3.75\times10^{4}/T$

Ea, for this reaction will be :

(1) 71.801×10^{4}

(2) 675928

(3) 536428

(4) 975001

CCK265

18. The time required for 10% completion of 1st order reaction at 298 is equal to that required to 25% completion at 318K. If value of A is $2.8\times10^{10}s^{-1}$. Calculate Ea.

(1) 39.91 KJ mol^{-1}

(2) 78.32 KJ mol^{-1}

(3) 30.07 KJ mol^{-1}

(4) 92.37 KJ mol^{-1}

CCK266

19. The initial concentration of N_2O_5 in 1st order reaction-

$N_2O_5(g)\longrightarrow 2NO_2(g)+\frac{1}{2}O_2(g)$ was $4\times10^{-2}M$ at 328K,

The concentration of N_2O_5 after 60 min was $0.2\times10^{-2}M$, Calculate rate constant (k) = ?

(1) 0.049 min^{-1}

(2) 0.375 min^{-1}

(3) 0.125 min^{-1}

(4) 0.278 min^{-1}

CCK267

20. In presence of catalyst, heat evolved or absorbed during reaction on -

(1) Decreases

(2) Increases

(3) Remains unchanged

(4) May increase or decrease

CCK268

21. Which of following statements is incorrect about the collision theory of chemical reaction ?

(1) The collision of atoms or molecules possessing sufficient threshold energy results into product formation.

(2) Molecules should collide with sufficient threshold energy and power.

(3) It considers reacting molecules or atom to be hard spheres and ignores their structural features.

(4) A number of effective collisions determine the rate of reaction.

CCK269

22. Rate Law can be determined from the balanced chemical equation if -

(1) The reverse reaction is involved

(2) Any of the reactant is in excess

(3) It is an elementary reactions

(4) It is a sequence of elementary reaction.

CCK270

23. An Elementary reaction is given below-

$A + B \longrightarrow 2P + 7Q$

Find the order of reaction

(1) 1 (2) 1.5 (3) 2 (4) zero

CCK271

24. Consider a reaction A ⟶ B with $K = 2.5 \times 10^{-2} s^{-1}$, calculate the concentration of 'A' remaining after 200s, if initial concentration of A is 1 mol L^{-1}

(1) 6.2×10^{-5} (2) 6.7×10^{-3}

(3) 5.3×10^{-2} (4) 4.7×10^{-2}

CCK272

25. Molecularity of a reaction can be.

(1) Zero (2) –2

(3) 1 (4) 0.5

CCK273

26. Activation energy is :

(1) Always equal to energy of product molecules

(2) The energy difference between activated complex and the energy of reactant molecules.

(3) Always Equal to energy of reactant molecules.

(4) Summation of threshold energy and energy of product molecules

CCK274

27. Consider the following statements

(a) Order of reaction is an experimental quantity

(b) Order of reaction can be zero or fraction.

(c) Order is applicable to elementary as well as complex reaction.

(d) For complex reaction, molecularity of the slowest step is same as the order of the overall reaction. If slowest step does not contain any intermediate.

The correct Statements is/are

(1) a, b and c only

(2) b and c only

(3) a and b only

(4) a, b, c and d

CCK275

28. The slope between log (a-x) vs t graph for 1st order reaction is :-

(1) –K

(2) –2.303 K

(3) $\frac{K}{2.303}$

(4) $-\frac{K}{2.303}$

CCK276

29. For 3rd order reaction, unit of rate constant (K) is given by :-

(1) $(\text{mol } L^{-1})^{-1}\ \text{Time}^{-1}$

(2) $(\text{mol } L^{-1})^{-2}\ \text{Time}^{-1}$

(3) $(\text{mol } L^{-1})^{0}\ \text{Time}^{-1}$

(4) $(\text{mol } L^{-1})^{2}\ \text{Time}^{-1}$

CCK277

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**1**	**2**	**1**	**2**	**4**	**2**	**3**	**2**	**1**	**2**	**1**	**4**	**3**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	**3**	**1**	**1**	**1**	**3**	**1**	**3**	**3**	**2**	**3**	**2**	**4**	**4**	**2**	

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